Tarachand Gupta | Lead AI Engineer

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AI engineering leader with **5+ years in full-stack development and AI product engineering**. Leads an **18-person team**, architecting and prototyping coding agents, context/retrieval systems, and developer tools; grounds implementation decisions in evaluation.

Experience

PureCode AI | Hyderabad / Remote

Technical Product Manager *Jan 2023 - Present*

- Architect and prototype systems for the **VS Code Assistant and Cloud Agent**; lead an **18-person team** across AI research, engineering, DevOps, and project management through technical decisions, developer UX, and **weekly fixes and fortnightly features**.
- Raised **internal task success from 82% to 90-95%** through **context engineering** over six months in early 2026; evaluate every new model and prompt change in **Langfuse with reasoning-model LLM judges**, updating judges as needed.
- Architected and prototyped **dynamic, specialized subagents** to parallelize independent tasks, keep the main context focused, and control token cost; designed a **shared scratchpad** for progress coordination.
- Shaped **hybrid retrieval, reranking, and repository maps** with embeddings, full-text search, and symbol exploration; architected and prototyped **file-based memory** with user/project scopes and on-demand recall across sessions.
- Improved token-cost efficiency by an **internally estimated 3x** through **prompt caching, stale-context pruning, and automatic summarization**.
- Designed **tool and terminal permissions and file protection** for developer control over actions, approvals, and protected files; prototyped the **MCP host** and guided clarification tools, reusable skills, and shared instructions.
- Guided cloud execution in **isolated environments**, with **checkpoints and validation-to-pull-request workflows** for reviewing and recovering agent changes.
- Built an **image-to-code ML pipeline** with **YOLOv7** UI detection, training-data collection/cleaning, and a domain-specific language (DSL); engineered a **DSL-to-code transpiler** across frontend frameworks, with theming, reusable components, and configurable code patterns.
- Helped build the **UI-generation platform and chat-based iteration**; guided **subscription, payment, and workspace systems** for teams.
- Product results: **monthly paid-subscriber retention rose from 65% to 70% over eight months**; **90% monthly enterprise customer retention**, excluding new customers.

Ovonts | Mumbai

Product Engineer *Nov 2021 - Jan 2023*

- Architected an **end-to-end campaign analytics and monitoring platform** connecting on-site events and influencer-content metrics for ROI analysis; designed **event-tracking SDKs, WordPress and e-commerce plugins**, and **scraping/data-collection pipelines** for Instagram and other platforms.
- Architected the **influencer discovery module** using deterministic rules, **probabilistic classification, and regression** to estimate demographics and profile attributes for campaign targeting.
- Built **RBAC with resource-level permissions and dynamic roles**, **API-key access**, multi-account/workspace systems, subscriptions, and session management.
- Maintained **AWS, Docker, and CI/CD**; adapted open-source tools, **reviewed code and mentored engineers**, and resolved architecture and delivery issues.

1st Main | Bengaluru

Full Stack Developer *Jul 2021 - Nov 2021*

- Built **React** interfaces with Material UI, Styled Components, and Ant Design; **Redux Toolkit/Redux-Saga** state and async flows; deployed **GCP/Firestore** APIs, **SendGrid** email services, and **Shopify GraphQL** integrations.

Skills

AI Systems: LLM applications, multi-agent orchestration, dynamic/specialized subagents, RAG, hybrid retrieval, embeddings, reranking, context engineering, prompt engineering/caching, context pruning/summarization, file-based memory, MCP host/client/server, agent skills, cloud agents, tool/terminal guardrails, Langfuse, LLM-as-a-judge, model/prompt evaluation.

Software Development: Python, Java, C++, TypeScript, JavaScript, HTML, CSS, React, Next.js, Vite, Node.js, Express, FastAPI, REST, GraphQL, WebSockets, message queues, Redux, Zustand, Tailwind CSS, Material UI, React Native, Expo.

Data & SaaS: PostgreSQL, SQLite, NoSQL/MongoDB, Redis, vector databases, application/database design, multi-tenant SaaS, RBAC, subscriptions, workspaces.

DevOps: AWS, Azure, GCP, Vercel, Docker, server management, GitHub Actions, CI/CD.

Technical Leadership: System design, solution engineering, prototyping, technical feasibility, user interviews, developer UX, user flows, roadmapping, feature scoping, code/design reviews, engineering mentorship, cross-functional leadership.

Projects

Subagent Bridge - Tauri/ACP cross-agent delegation with model rules and stateful follow-ups. Web GitHub

Bookmark AI - Hybrid SQLite/vector search, Redis live-tab sync, and a BYOK AI assistant. Web GitHub

God MCP - Python/FastMCP tool-server generation and selected-host registration. Web GitHub

Project Orbit AI - AI-spawned 3D objects, schema validation, interaction APIs, and Rapier physics. Web GitHub

Education

Bachelor of Engineering, Universal College of Engineering, Mumbai | **7.8 CGPI** | **June 2021**
Diploma in Information Technology, Pravin Patil College of Diploma, Mumbai